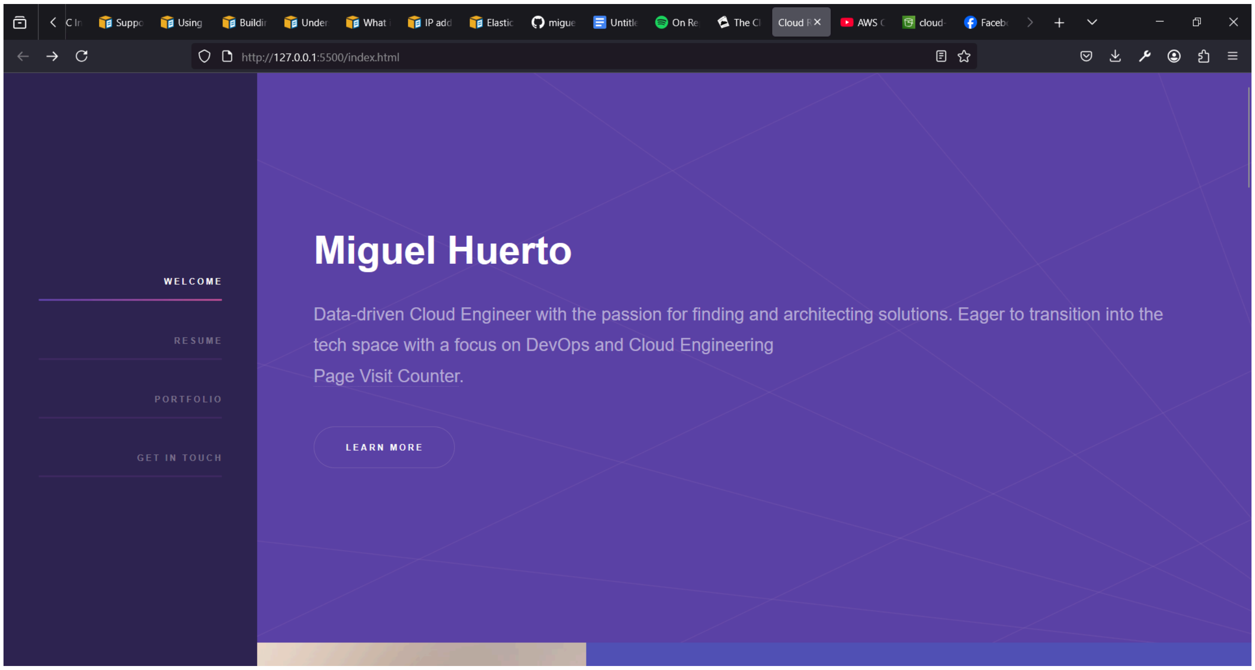
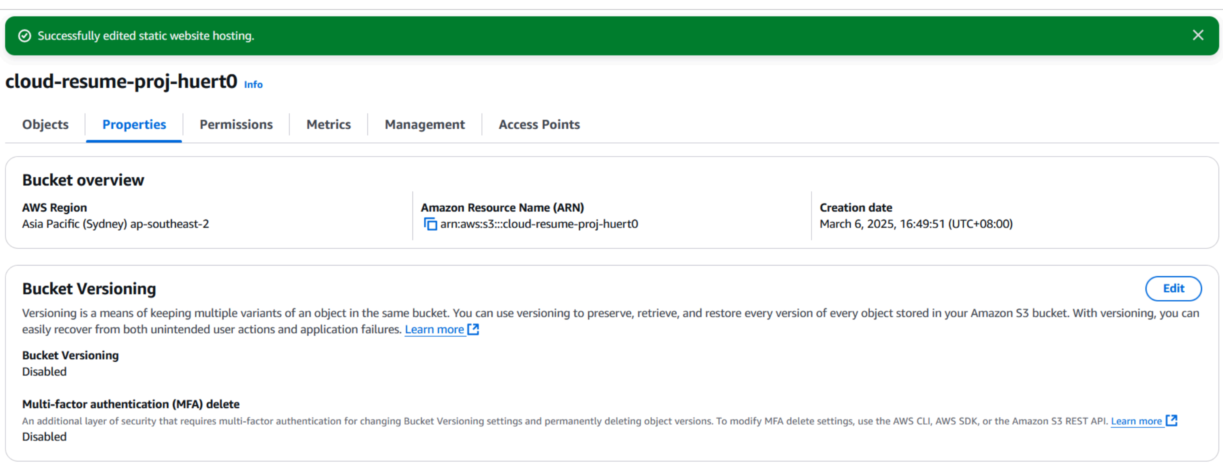
AWS Cloud Resume Challenge

This project will be composed of several parts

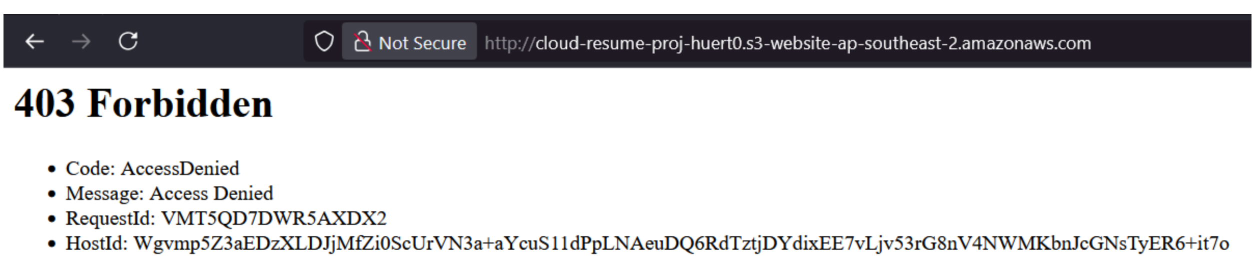
1. **HTML&CSS**



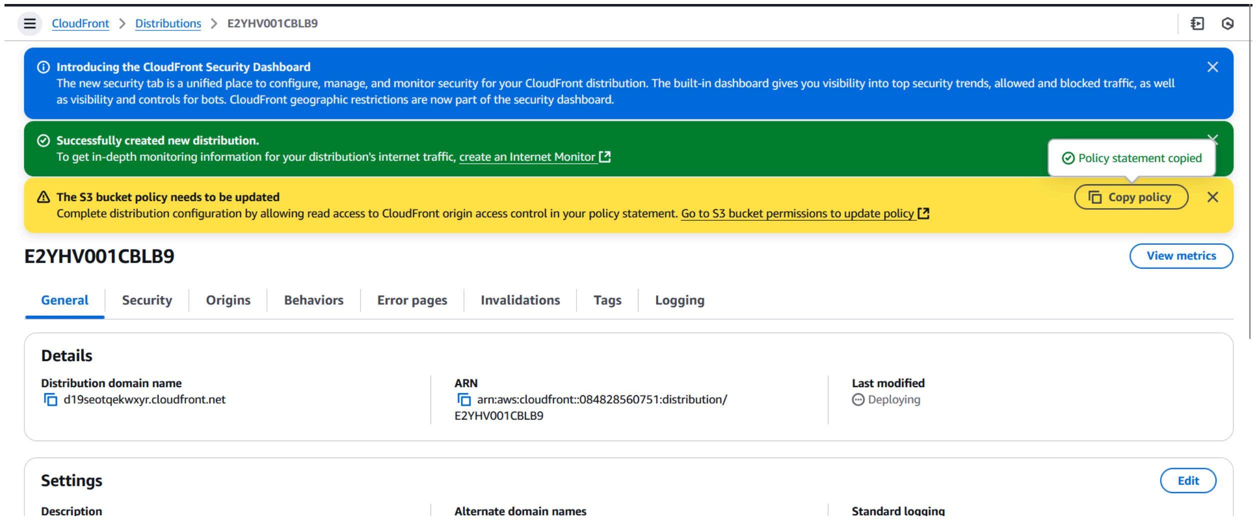
2. **Static Website**



Public Access Blocked



Use CloudFront for low latency high transfer speed delivery of data, CDN.

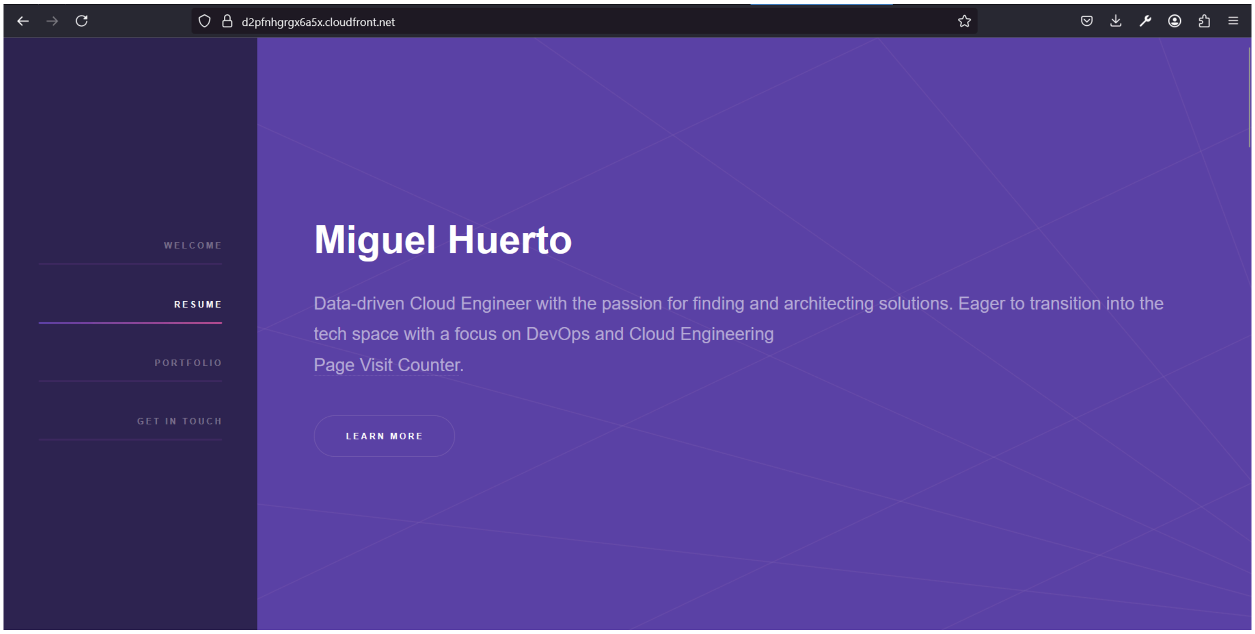


HTTPS only

Edited Permissions

```
{
  "Version": "2008-10-17",
  "Id": "PolicyForCloudFrontPrivateContent",
  "Statement": [
    {
      "Sid": "AllowCloudFrontServicePrincipal",
      "Effect": "Allow",
      "Principal": {
        "Service": "cloudfront.amazonaws.com"
      },
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::cloud-resume-proj-huert0/*",
      "Condition": {
        "StringEquals": {
          "AWS:SourceArn": "arn:aws:cloudfront::084828560751:distribution/E2YHV001CBLB9"
        }
      }
    }
  ]
}
```

3. **HTTPS**



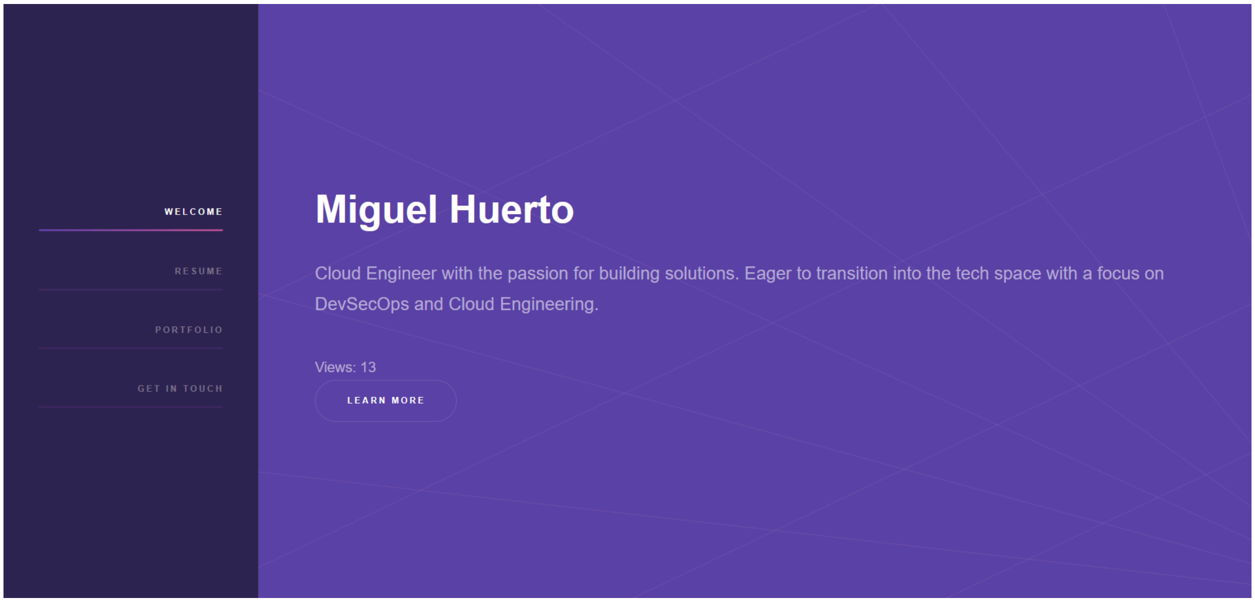
4. **DNS**

5. **Javascript**

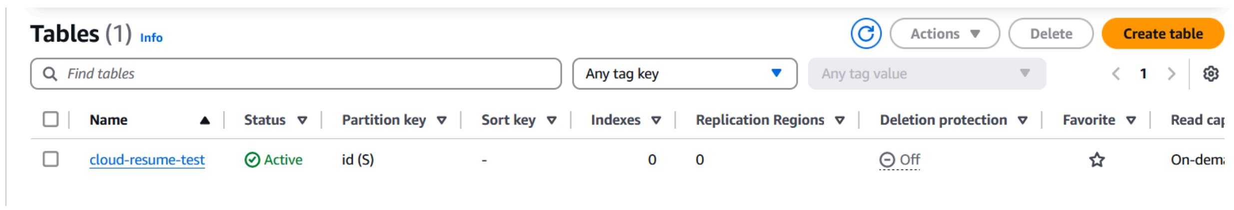
index.html | index.js

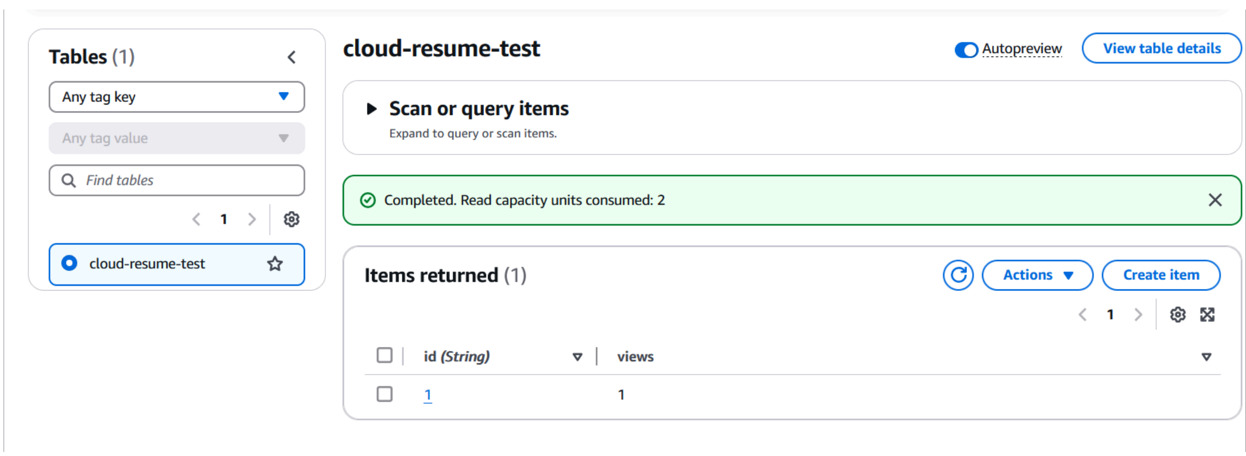
assets > js > index.js > updateCounter

```javascript
const counter = document.querySelector(".counter-number");

async function updateCounter() {
    let response = await fetch("https://nrtu4vkqkncjk3huxyfxwjn47m0throo.lambda-url.ap-southeast-2.on.aws/");
    let data = await response.json();
    counter.innerHTML = `Views: ${data.views}`;
}

updateCounter();
```

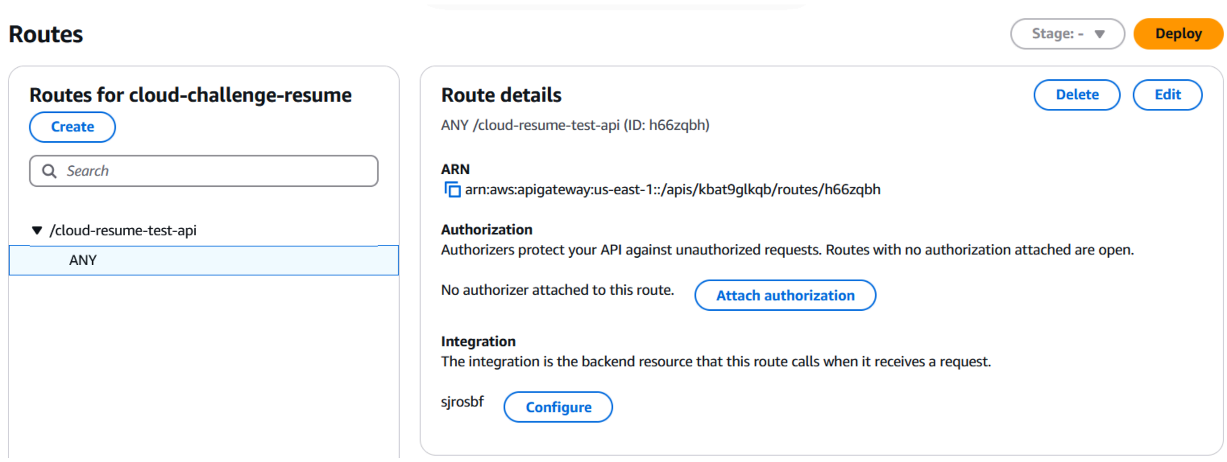


6. Database





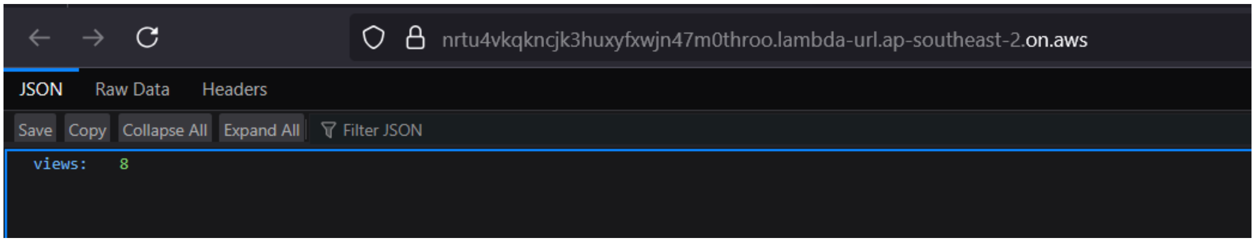
7. API



8. **Python**

lambda_function.py

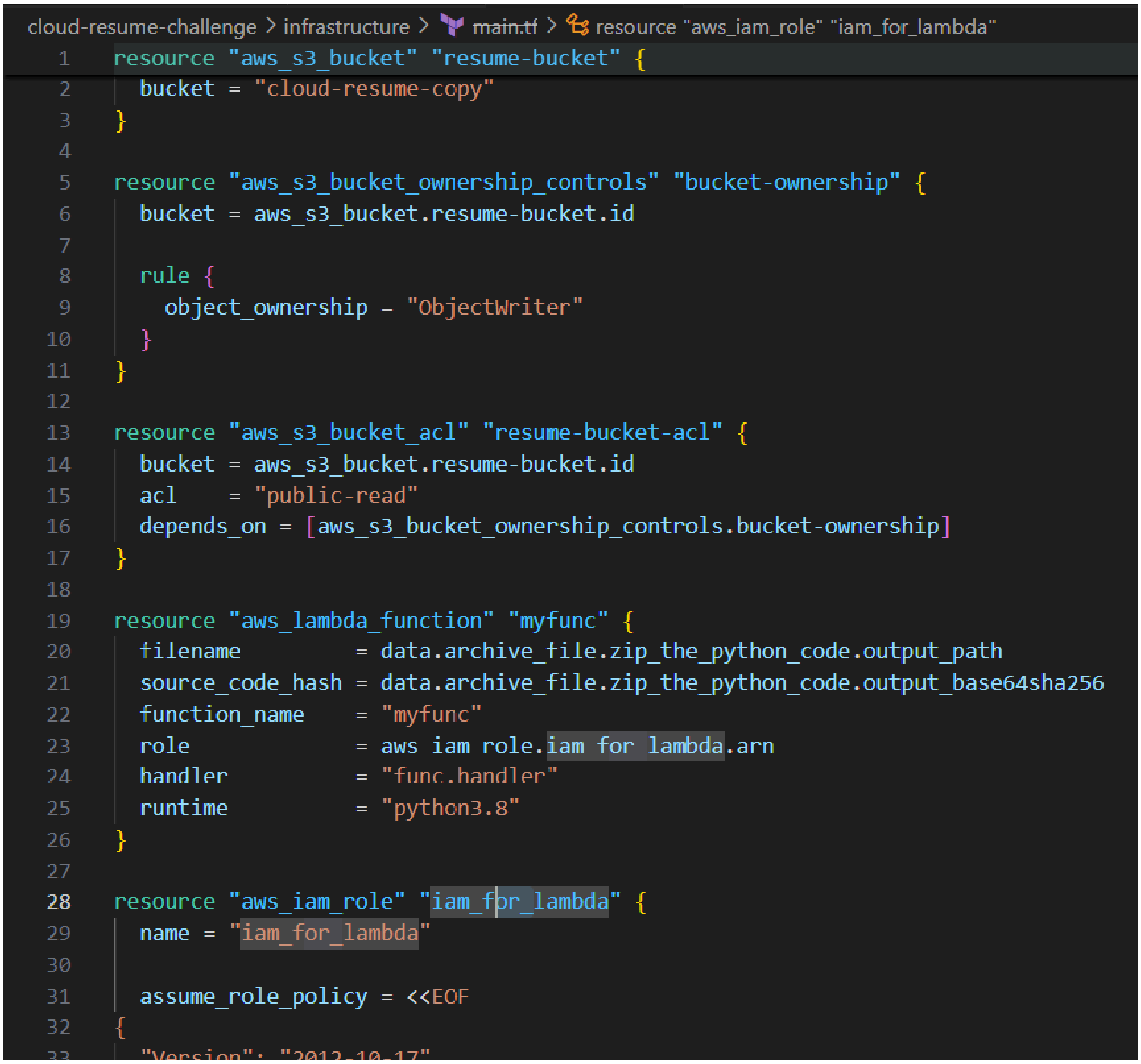
```python
import json
import boto3

dynamodb = boto3.resource('dynamodb')
table = dynamodb.Table('cloud-resume-test')

def lambda_handler(event, context):
    response = table.get_item(Key={'id': '0'})
    views = response['Item']['views']
    views += 1

    table.update_item(
        Key={'id': '0'},
        UpdateExpression="SET #v = :val",
        ExpressionAttributeNames={"#v": "views"},
        ExpressionAttributeValues={":val": views},
        ReturnValues="UPDATED_NEW"
    )

    return {"views": views}

```



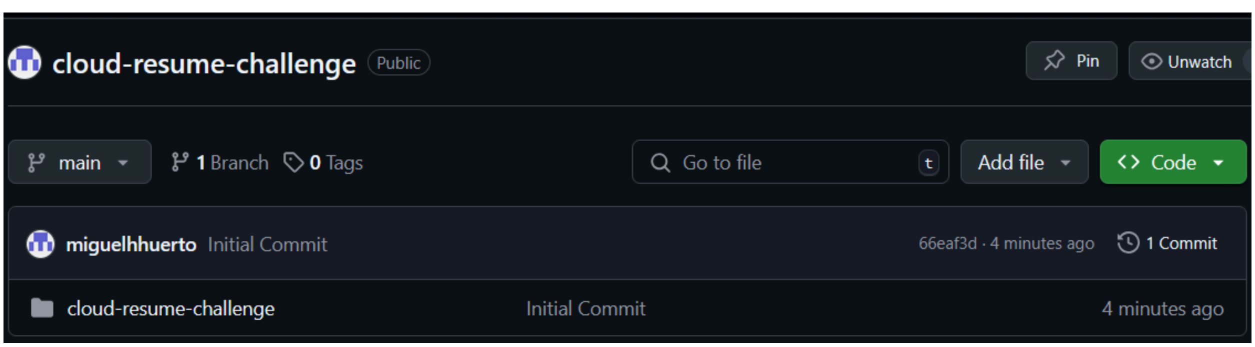
9. **Infrastructure as Code (TerraForm)**

cloud-resume-challenge > infrastructure > provider.tf > ...

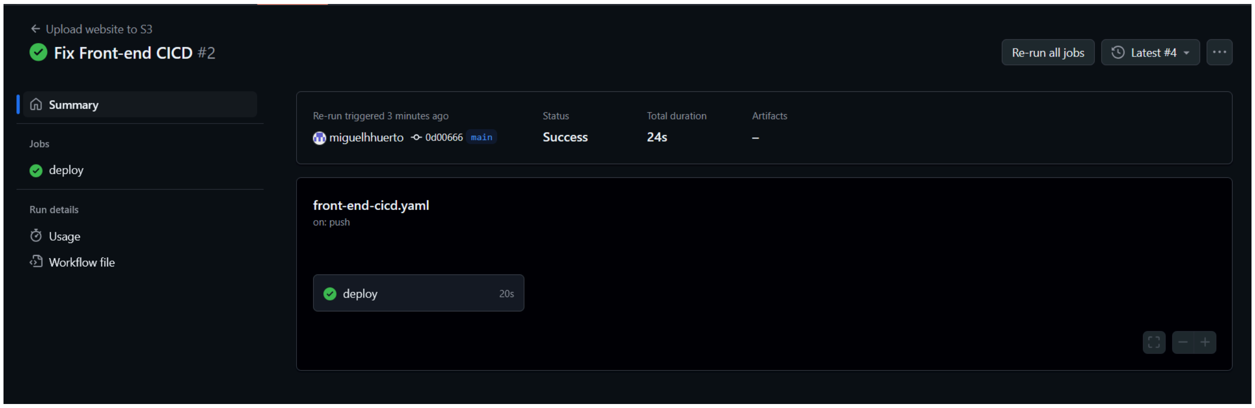
```
terraform {
  required_providers {
    aws = {
      source  = "hashicorp/aws"
      version = "~> 5.0"
    }
  }
}

provider "aws" {
  region = "ap-southeast-2"
  profile = "default"
}

```

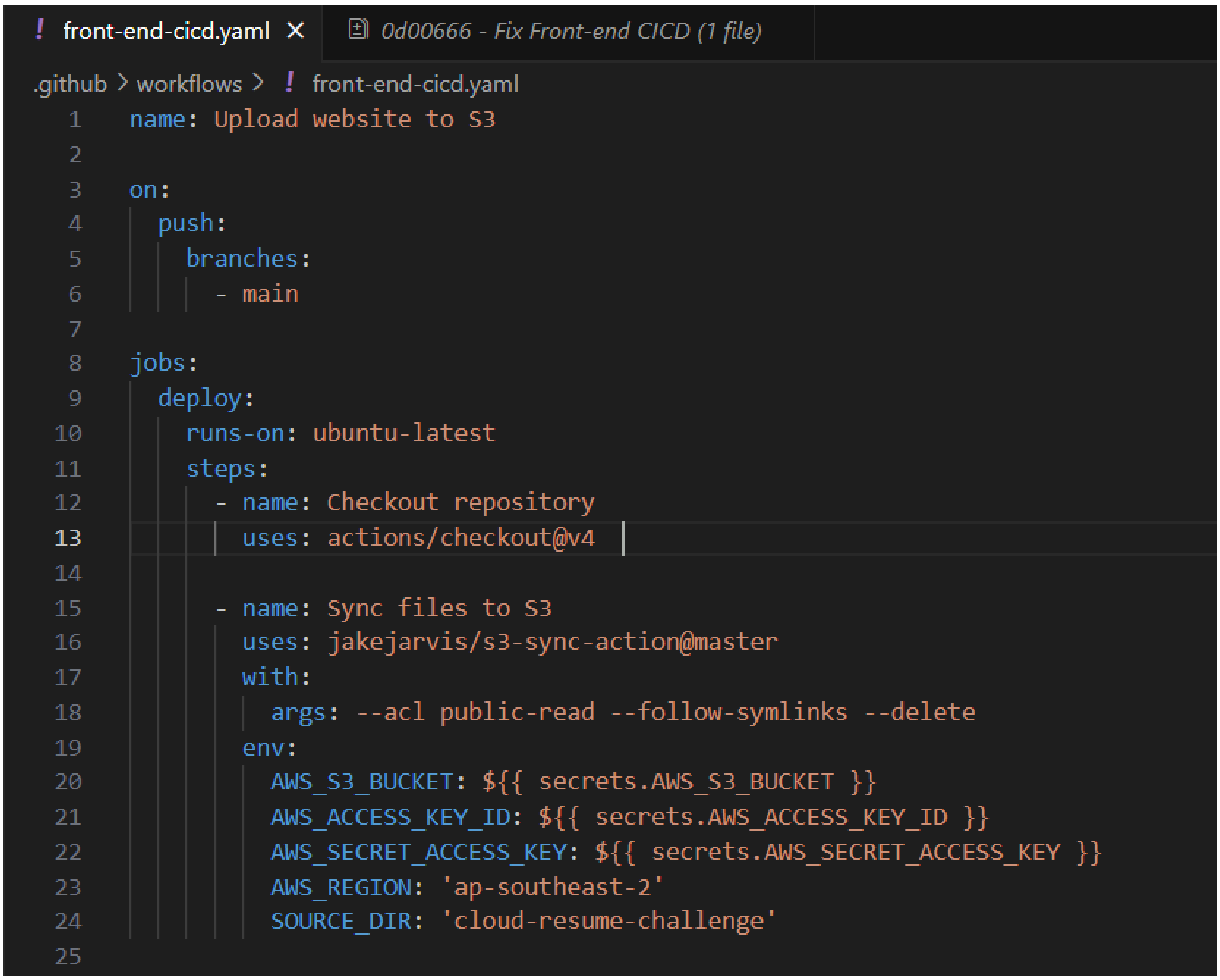


10. Source Control



11. CI/CD (GitHub Actions)






```yaml
name: Upload website to S3

on:
  push:
    branches:
      - main

jobs:
  deploy:
    runs-on: ubuntu-latest
    steps:
      - name: Checkout repository
        uses: actions/checkout@v4

      - name: Sync files to S3
        uses: jakejarvis/s3-sync-action@master
        with:
          args: --acl public-read --follow-symlinks --delete
        env:
          AWS_S3_BUCKET: ${{ secrets.AWS_S3_BUCKET }}
          AWS_ACCESS_KEY_ID: ${{ secrets.AWS_ACCESS_KEY_ID }}
          AWS_SECRET_ACCESS_KEY: ${{ secrets.AWS_SECRET_ACCESS_KEY }}
          AWS_REGION: 'ap-southeast-2'
          SOURCE_DIR: 'cloud-resume-challenge'
```